Question 25

25 In order to keep the store properly stocked with produce, a grocery store manager wants to determine the most popular day of the week for people to buy apples and oranges. She randomly collects a sample of sales records for each day for three consecutive weeks. Is this type of data collection method a survey, an observational study, or an experiment? Explain your reasoning.

This study is an observational study because she is not changing any variables.

Question 26

26 Given the following functions $f(x) = 2x^2 + 3x - 7$ and $g(x) = 2x - 5$, determine $f(x) - g(x)$, and express the result in standard form.

$2x^2 + 3x - 7 - 2x - 5$

$2x^2 + 1x - 12$

Question 27

27 Determine algebraically the zeros of $f(x) = x^3 - 4x^2 - 9x + 36$.

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ $a = 1$ $b = -4$

	x	-4
x^2	x^3	$-4x^2$
-9	$-9x$	$+36$

$(x-4)(x^2-9)$

$(x-4)(x-3)(x+3)$

$x+4 \quad x-3 \quad x+3$

$\{-4, +3, -3\}$

Question 29

29 Express $\left(-2 + \sqrt{-25}\right)^2$ in simplest $a + bi$ form. Justify your answer.

$(-2 + \sqrt{-25})(-2 + \sqrt{-25})$

$4 - 2\sqrt{-25} - 2\sqrt{-25} - 25$

$-4\sqrt{-25} - 21$

$-4i(5) - 21$

$-21 - 20i$

$21 + 20i$

Question 30

30 The most common radioactive form of cesium is cesium-137. The half-life of cesium-137 is 30 years.

Suppose 2.7 micrograms of cesium-137 are found in a soil sample. Write a function for the amount of cesium-137 remaining, $c(t)$, in micrograms, in terms of the number of years, t, since the soil sample was taken.

$y = 2.7\left(\frac{1}{2}\right)^{\frac{x}{30}}$

Question 31

31 Solve algebraically for x:

$$\sum_{a=0}^{4} (ax - 3) = 5$$

$(0x-3) = 5$

$(1x-3) = 5$

$(2x-3) = 5$

$(3x-3) = 5$

$(4x-3) = 5$

$+$

$(10x-15) = 25$

$+15 \quad +15$

$\frac{10x = 40}{10} \quad x = 4$

Question 32

32 Given the exponential function $y = 5.381e^{0.214x}$, rewrite the function in the form of $y = a(b)^x$, rounding values to the *nearest thousandth*.

$y = 5.381(e)^{0.214x}$

$y = 14.627^{0.214x}$

$y = 1.778^x$

$y = 5.381 \cdot 1.239^x$

Question 33

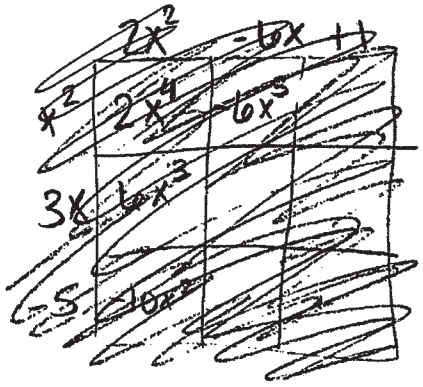
33 Determine the quotient and remainder of the expression below.

$$\frac{2x^4 - 27x^2 + 33x - 8}{x^2 + 3x - 5}$$

Express the result in the form $q(x) + \frac{r(x)}{d(x)}$.

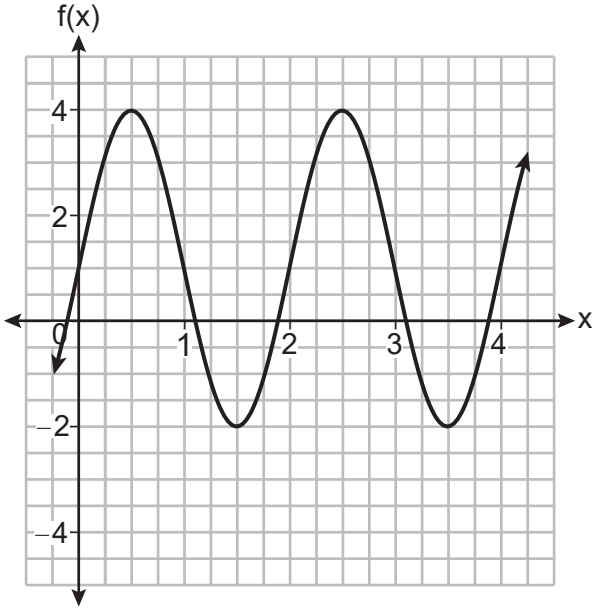
$$\begin{array}{r} 2x^2 - 6x + 1 \\ x^2+3x-5 \overline{)\,2x^4 + 0x^3 - 27x^2 + 33x - 8} \\ +\,\cancel{-2x^4 \mp 6x^3 + 10x^2} \\ \hline -6x^3 - 17x^2 + 33x \\ +\,\cancel{+6x^3 + 18x^2 + 30x} \\ \hline x^2 + 63x - 8 \\ +\,\cancel{-x^2 \mp 3x + 5} \\ \hline 60x - 3 \end{array}$$

$$2x^2 - 6x + 1 + \frac{60x - 3}{x^2 + 3x - 5}$$



Question 34

34 The sinusoidal function f is graphed below.



State the period of the function f.

P = 2.5

State the amplitude of the function f.

A = 3

$A = \frac{max - min}{2}$

$A = \frac{4 - -2}{2} = 3$

Write the equation in the form $f(x) = A \sin(Bx) + D$ to represent the sinusoidal function.

$f(x) = 3\sin(0.8\pi x) + 1$

$A = 3$

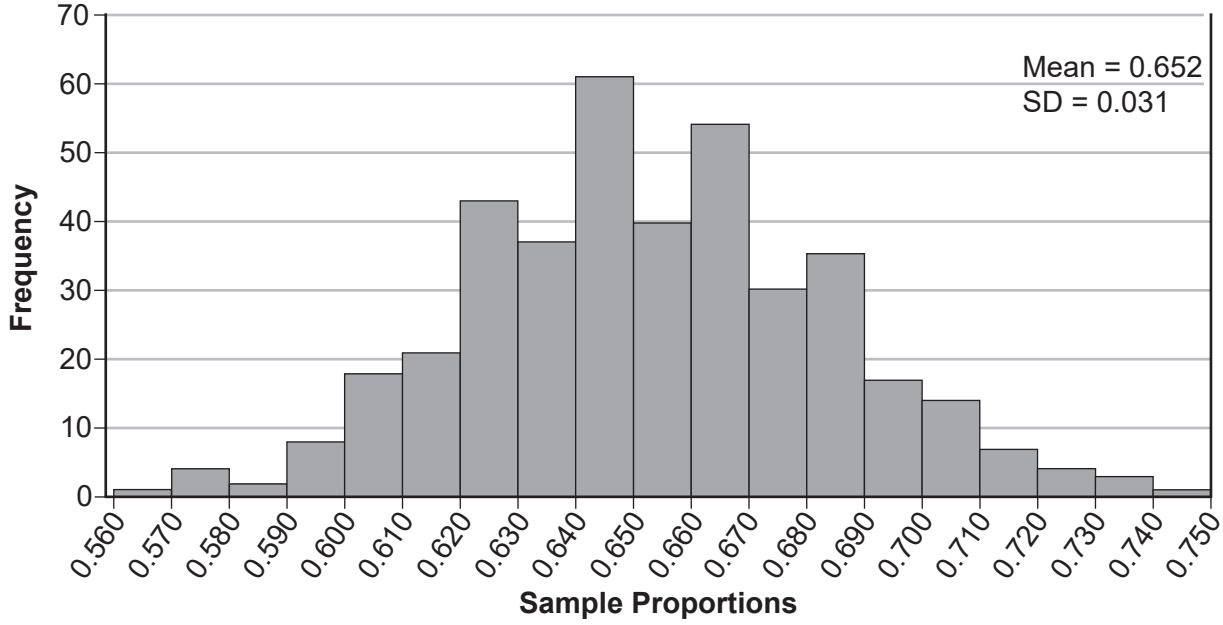
$D = \frac{max + min}{2} = \frac{4 + -2}{2} = 1$

$D = 1$

$B = \frac{2\pi}{2.5} = 0.8\pi$

Question 35

35 A principal wonders what percentage of parents at her large high school restrict their child's cell phone privileges. The school surveys a random sample of 250 parents at the school and finds that 65% of them restrict their child's cell phone privileges. The school runs a simulation with 400 samples of 250 parents, assuming that 65% of them restrict cell phone privileges. The simulation results are shown below.



Use the simulation to determine an interval containing the middle 95% of simulated proportions. Justify your answer. Round all values to the *nearest hundredth*.

.59 to .71

The school district's superintendent worries that less than half of parents restrict their child's cell phone privileges. Do the simulation results support the superintendent's concern? Explain.

No. The plausible outcomes are all more than 50%, so it would be statistically significant for less than ½ to restrict.

Question 36

36 Lubin pours green tea into a cup to cool. The function below can be used to model the temperature, in degrees Fahrenheit, of Lubin's tea x minutes after he pours it into the cup.

$$T(x) = 110e^{-0.0453x} + 74$$

State the temperature of the tea, in degrees Fahrenheit, when Lubin poured it into the cup.

~~74°F~~ 184°F

To the *nearest tenth*, use $T(x)$ to determine the average rate of change in degrees per minute, from $x = 0$ to $x = 5$.

$$\frac{161.705 - 184}{5 - 0} \qquad \frac{-22.295}{5}$$

$$-4.5$$

Question 36 is continued on the next page.

Question 36 continued

Interpret the value of the average rate of change within the given context.

4.5°F less every minute the cup is cooled

To the *nearest tenth of a minute*, determine how long Lubin must wait after pouring his tea into the cup before the tea's temperature reaches 120° Fahrenheit.

$120 = 110e^{-.0453x} + 74$

$\ln \frac{46}{110} = -.0453x$

$\frac{\ln \frac{46}{110}}{-.0453}$

19.2 minutes